MCQ QUESTIONS

Quantum Computing

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Quantum Computing - Multiple-Choice Questions (Tertiary Level)

Each question has four options (A-D), the correct answer, and a concise explanation.

Q1. [Difficulty: Easy]
Which of the following best describes a qubit?

A) A classical bit that can be either 0 or 1.

B) A two-level quantum system that can exist in a superposition $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

C) A photon that always travels at the speed of light.

D) A logical variable used in Boolean algebra.

✔ Correct Answer: B

Explanation: A qubit is a quantum two-level system whose state is a normalized linear combination of the computational basis states $|0\rangle$ and $|1\rangle$. Options A, C and D describe classical or unrelated concepts.

Q2. [Difficulty: Easy]
What does the No-Cloning Theorem forbid?

A) Measuring a qubit in the computational basis.

B) Creating an exact copy of an arbitrary unknown quantum state.

C) Entangling two qubits that are far apart.

D) Applying a unitary gate to a single qubit.

✔ Correct Answer: B

Explanation: The theorem states that no physical operation can duplicate an arbitrary unknown quantum state perfectly. Measurement, entanglement, and unitary gates are all allowed operations.

Q3. [Difficulty: Easy]
Which gate is the quantum analogue of the classical NOT operation?

A) H (Hadamard) gate

B) X gate

C) S gate

D) T gate

✔ Correct Answer: B

Explanation: The X (Pauli-X) gate flips $|0\rangle \leftrightarrow |1\rangle$, directly mirroring the classical NOT. H creates superposition; S and T are phase-shift gates.

Q4. [Difficulty: Medium]
Consider a 3-qubit register initialized to |000■. After applying a Hadamard gate to each qubit, what is the probability of measuring the state |101■?

A) 0

B) 1/8

C) 1/4

D) 1/2

✔ Correct Answer: B

Explanation: Each Hadamard creates an equal superposition of |0■ and |1■, yielding $2^3 = 8$ equally likely basis states. Hence any specific outcome, including |101■, occurs with probability 1/8.

Q5. [Difficulty: Medium]
Which of the following statements about quantum entanglement is FALSE?

A) Entangled states cannot be written as a tensor product of individual qubit states.

B) Entanglement can be used as a resource for quantum teleportation.

C) Measuring one qubit of an entangled pair always leaves the other qubit in a pure state independent of the measurement outcome.

D) Entanglement entropy quantifies the amount of correlation between subsystems.

✔ Correct Answer: C

Explanation: After measuring one qubit, the post-measurement state of the partner depends on the measurement result; it is not independent. Statements A, B, and D are true.

Q6. [Difficulty: Medium]
In the context of quantum error correction, the [[7,1,3]] Steane code encodes how many logical qubits into how many physical qubits, and what is its distance?

A) 1 logical qubit, 7 physical qubits, distance 3.

B) 7 logical qubits, 1 physical qubit, distance 3.

C) 1 logical qubit, 7 physical qubits, distance 7.

D) 3 logical qubits, 7 physical qubits, distance 1.

✔ Correct Answer: A

Explanation: The notation [[n,k,d]] means *n* physical qubits, *k* logical qubits, and code distance *d*. Thus the Steane code is [[7,1,3]].

Q7. [Difficulty: Medium]
Which complexity class captures decision problems efficiently solvable by a quantum computer with bounded error probability?

A) NP

B) BQP

C) PSPACE

D) QMA

✔ Correct Answer: B

Explanation: BQP (Bounded-Error Quantum Polynomial time) is the quantum analogue of BPP, defining problems solvable in polynomial time with error ≤ 1/3. NP and PSPACE are classical classes; QMA is the quantum analogue of NP with a quantum proof.

Q8. [Difficulty: Hard]
A quantum circuit on two qubits performs: (1) H on qubit 1, (2) CNOT (control 1 -> target 2), (3) S on qubit 2, (4) measurement of both qubits in the computational basis. Which statement about the measurement statistics is correct?

A) The outcomes are uniformly distributed over the four basis states.

B) The probability of obtaining |00■ equals the probability of obtaining |11■, and both are 1/2.

C) The outcome |01■ never occurs.

D) The probability of |10■ is larger than that of |01■ because of the S gate.

✔ Correct Answer: C

Explanation: Steps (1)-(2) generate the Bell state (|00■ + |11■)/√2. The S gate adds a relative phase to the |11■ component but does not affect measurement probabilities. Consequently only |00■ and |11■ can be observed; |01■ (and |10■) never appear.

Q9. [Difficulty: Hard]
In a surface-code implementation, the logical error rate per round is approximated by $$ p_{\text{L}} \approx p \,\frac{d}{2}\,(0.1)^{(d-1)/2}, $$ where *p* is the physical gate error probability and *d* the code distance. If *p* = 10■³, what is the smallest integer *d* that yields *p*■ < 10■■?

A) d = 3

B) d = 5

C) d = 7

D) d = 9

✔ Correct Answer: D

Explanation: Evaluate the expression: d = 5 -> p■ ~ 10■³ · 2.5 · 0.01 = 2.5 × 10■■ (too large) d = 7 -> p■ ~ 10■³ · 3.5 · 0.001 = 3.5 × 10■■ (still > 10■■) d = 9 -> p■ ~ 10■³ · 4.5 · 0.0001 = 4.5 × 10■■ (< 10■■) Thus *d* = 9 is the smallest distance meeting the requirement.

Q10. [Difficulty: Hard]
Shor's algorithm relies on quantum phase estimation (QPE) as a subroutine. Which statement about QPE is TRUE?

A) QPE can estimate the eigenvalue of any Hermitian operator without prior knowledge of the eigenstate.

B) The precision of QPE scales linearly with the number of ancilla qubits used.

C) QPE requires measurement in the computational basis only after all controlled-unitary operations are applied.

D) QPE is ineffective for finding the order of an integer modulo *N* because it cannot handle periodicity.

✔ Correct Answer: C

Explanation: In QPE the ancilla register undergoes a series of controlled-U^{2^k} operations, followed by an inverse QFT, and finally a single computational-basis measurement. Option A is false because the eigenstate must be supplied; B is false because precision improves exponentially (∝ 2■); D contradicts the core use of QPE in order-finding.

Q11. [Difficulty: Medium]
Which physical implementation of qubits typically offers the longest intrinsic coherence times when environmental isolation is comparable?

A) Superconducting transmon qubits

B) Trapped-ion hyperfine qubits

C) Photonic time-bin qubits

D) Topological Majorana zero modes

✔ Correct Answer: B

Explanation: Trapped-ion hyperfine qubits routinely achieve coherence times ranging from seconds to minutes, far exceeding the micro- to millisecond coherence of superconducting qubits, the loss-limited coherence of photonic qubits, and the still-experimental nature of topological qubits.

Q12. [Difficulty: Medium]
Which of the following best characterizes the known relationship between BQP and PSPACE?

A) BQP ⊂ PSPACE, and it is known that BQP ≠ PSPACE.

B) BQP = PSPACE, proven by quantum simulation of classical space-bounded machines.

C) BQP ⊆ PSPACE, but whether the inclusion is strict remains open.

D) PSPACE ⊂ BQP, because quantum computers can solve all PSPACE problems efficiently.

✔ Correct Answer: C

Explanation: Any quantum computation that runs in polynomial time can be simulated using polynomial space, giving the inclusion BQP ⊆ PSPACE. Whether this inclusion is strict (i.e., BQP ≠ PSPACE) is an open problem in complexity theory. Options A, B, and D assert relationships that have not been proven.

Q13. [Difficulty: Easy]
What is the primary purpose of the measurement step in a quantum algorithm?

A) To increase entanglement among qubits.

B) To collapse the quantum state and obtain classical output bits.

C) To apply a unitary transformation to the register.

D) To reset all qubits to the |0■ state.

✔ Correct Answer: B

Explanation: Measurement projects the quantum state onto a basis, producing classical outcomes and destroying superposition. It does not create entanglement, is not a unitary operation, and does not necessarily reset qubits.

Q14. [Difficulty: Medium]
In the variational quantum eigensolver (VQE), which component is performed classically?

A) Preparation of the parameterized quantum ansatz.

B) Execution of the quantum circuit on hardware.

C) Evaluation of the expectation value of the Hamiltonian.

D) Optimization of the variational parameters using a classical optimizer.

✔ Correct Answer: D

Explanation: VQE is a hybrid algorithm: a classical optimizer updates the parameters based on measured expectation values, while the quantum processor prepares the ansatz, runs the circuit, and measures the expectation values.

Q15. [Difficulty: Hard]
Consider two single-qubit quantum channels ■■ and ■■ with Kraus sets {K*i} and {L*j}, respectively. Their composition ■ = ■■ ■ ■■ is also a quantum channel. Which statement about the Choi matrix J(■) of the composite channel is always true?

A) J(■) = J(■■) ⊗ J(■■).

B) The eigenvalues of J(■) are the products of the eigenvalues of J(■■) and J(■■).

C) J(■) is positive semidefinite and its trace equals the dimension of the input space.

D) J(■) is unitary if both ■■ and ■■ are unitary channels.

✔ Correct Answer: C

Explanation: The Choi matrix of any completely positive, trace-preserving (CPTP) map is always positive semidefinite, and its trace equals the dimension of the input Hilbert space (here, 2 for a qubit). Tensoring (A) and eigenvalue multiplication (B) are not generally valid for channel composition, and while unitary channels have rank-1 Choi matrices (projectors), they are not unitary operators, so D is false.